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Original Article

Tobacco Smoke Exposure in Early Childhood and Later Risk of Inflammatory Bowel Disease: A Scandinavian Birth Cohort Study

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Abstract

Objectives: To examine the association between early-life smoking exposure and later risk of inflammatory bowel disease [IBD].

Methods: We followed 115663 participants from the Norwegian Mother, Father and Child [MoBa] and All Babies in Southeast Sweden [ABIS] cohorts from birth [1997–2009] through 2021. IBD was identified through national patient registers. Validated questionnaire data defined maternal smoking during pregnancy, maternal environmental tobacco smoke [ETS] exposure during pregnancy, and child ETS exposure by ages 12 and 36 months. Cox regression was used to estimate adjusted hazard ratios [aHRs] for sex, maternal age, education level, parental IBD, and origin. Cohort-specific estimates were pooled using a random-effects model.

Results: During 1 987 430 person-years of follow-up, 444 participants developed IBD [ABIS, 112; MoBa, 332]. Any vs no maternal smoking during pregnancy yielded a pooled aHR of 1.30 [95% CI = 0.97–1.74] for offspring IBD. Higher level of maternal smoking during pregnancy (compared with no smoking, average ≥6 cigarettes/day: pooled aHR = 1.60 [95% CI = 1.08–2.38]) was associated with offspring IBD, whereas a lower smoking level was not (average 1–5 cigarettes/day: pooled aHR = 1.09 [95% CI = 0.73–1.64]). Child ETS exposure in the first year of life was associated with later IBD (any vs no ETS, pooled aHR = 1.32 [95% CI = 1.03–1.69]). Estimates observed for child ETS exposure by 36 months were similar but not statistically significant.

Conclusions: In this prospective Scandinavian cohort study, children exposed to higher levels of maternal smoking during pregnancy or ETS during the first year of life were at increased risk of later IBD.

Key Words: Inflammatory bowel disease; smoking exposure; childhood; ABIS; MoBa

1. Introduction

Inflammatory bowel disease [IBD] is a chronic, immune-mediated disease that includes two main subtypes: Crohn's disease [CD] and ulcerative colitis [UC]. If the subtype cannot be specified, it is referred to as IBD-unclassified [IBD-U]. The incidence of IBD is increasing worldwide and is exeptionally high in Northern Europe.[1–3]

Evidence suggests early-life environmental factors may influence the development of IBD,[4,5] as genetic changes are too slow to explain the rise in IBD incidence observed in the past decades. Current smoking in adults is consistently associated with an increased risk of later CD and reduced risk of UC.[6] Early-life smoking exposure has been associated with later IBD,[7] but such data have with few exceptions[8–10] been based on retrospective reports [Supplementary Table 1] liable to recall bias or even reverse causation.

Although the exact mechanisms by which tobacco smoke can affect the risk of IBD are not understood, tobacco smoke has several adverse effects. Tobacco smoke has been linked to both anti- and pro-inflammatory pathways[6,11] and dysfunction of the gastrointestinal mucosa.[12] During pregnancy, smoking has been associated with altered gut microbiota[13] and differential DNA methylation in the child,[14] possibly contributing to IBD development.

Based on prospectively collected data from two Scandinavian birth cohorts, we investigated the association between early-life smoking exposure and the later occurrence of IBD, to test the hypothesis that early-life environmental tobacco smoke [ETS] exposure may influence the risk of IBD.

2. Methods

This study was based on the prospective birth cohort studies All Babies in Southeast Sweden [ABIS] and the Norwegian Mother, Father, and Child Cohort [MoBa] [Supplementary Figure 1].

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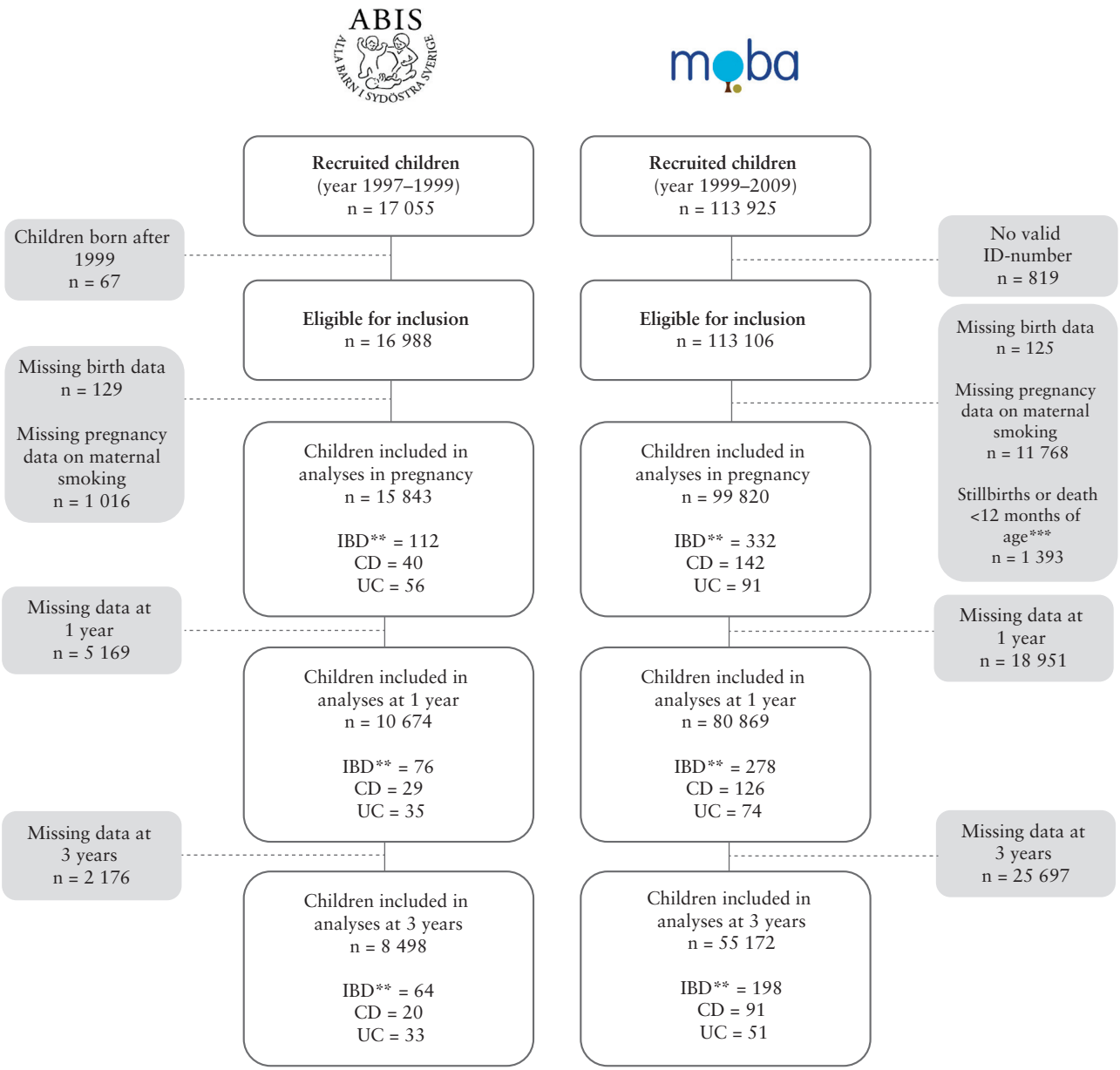


Figure 1 Flowchart of study participants in the ABIS [participation rate 79%] and MoBa [participation rate 41%] birth cohorts. ABIS, All Babies in Southeast Sweden study; CD, Crohn's disease; IBD, inflammatory bowel disease; MoBa, the Norwegian Mother, Father, and Child Cohort Study; UC, ulcerative colitis. *Other reason: MoBa, unknown reason. **Containing any subtype of inflammatory bowel disease [IBD], including IBD-unclassified. ***Data on stillbirths or deaths <12 months of age were not available in ABIS.

ABIS is a birth cohort study inviting parents to children born in Southeast Sweden from October 1, 1997, to October 1, 1999, to participate. About 21 700 children were born during this period. Some 17 055 children were included in ABIS [participation rate 79%]. The current study was restricted to 15 843 participants with data on maternal smoking exposure during pregnancy. MoBa is a birth cohort conducted by the Norwegian Institute of Public Health. Some 113 925 children were included between 1999 and 2009 [participation rate 41%].[15] The present study was restricted to 99 820 MoBa participants with data on maternal smoking during pregnancy [Figure 1].

2.1. Exposures: measures of early-life smoking exposure

Maternal smoking during pregnancy was reported in the at-birth questionnaire in ABIS and in MoBa questionnaires administered at 15 and 30 weeks of pregnancy and at the child's age of 6 months [covering smoking habits after 30 weeks of pregnancy]. In this study, smoking during pregnancy was modelled as a binary [yes/no] and trichotomous variable of the average daily cigarettes smoked per day during pregnancy (no smoking [reference], smoking 1–5 cigarettes/day [including mothers smoking on average less than one full cigarette per day of pregnancy in MoBa] and smoking ≥6 cigarettes/day; Supplementary Table 2). Self-reported maternal smoking during pregnancy has been validated in both ABIS and MoBa against measures of the child's cotinine levels, a biomarker of nicotine exposure.[16–19] Inconsistencies in MoBa across questions related to ever smoking and any smoking during pregnancy were found in 116 children, and the mothers of 78 children reported inconsistencies for ever smoking and the level of smoking during pregnancy [number of cigarettes].

Table 1 Participant characteristics in the ABIS and MoBa birth cohorts according to maternal smoking during pregnancy.

		ABIS		MoBa		
	All 15 843	No smoking during pregnancy 14 083	Smoking during pregnancy 1 760	All 99 820	No smoking during pregnancy 90 128	Smoking during pregnancy 9 692
Sex [%]						
Female	7 659 [48.3]	6792 [48.2]	867 [49.3]	48 668 [48.8]	43 919 [48.7]	4749 [49.0]
Male	8 184 [51.7]	7291 [51.8]	893 [50.7]	51,152 [51.2]	46 209 [51.3]	4943 [51.0]
Age at the end of follow-up[a] [years]						
Mean [SD]	22.2 [1.0]	22.2 [0.9]	22.2 [1.5]	16.4 [2.1]	16.3 [2.1]	17.2 [2.2]
Median [interquartile range]	22.3 [21.8; 22.8]	22.3 [21.8; 22.8]	22.3 [21.8; 22.8]	16.2 [14.7; 18.0]	16.1 [14.6; 17.9]	17.2 [15.4; 18.8]
Age at IBD diagnosis [years]	112	99	13	332	285	47
Mean [SD]	16.9 [3.7]	16.9 [3.7]	17.1 [3.7]	12.7 [3.7]	12.7 [3.7]	13.0 [3.9]
Median [interquartile range]	17.9 [14.9; 19.6]	17.9 [14.7; 19.6]	18.0 [16.3; 19.3]	13.2 [10.7; 15.5]	13.2 [10.7; 15.3]	13.0 [10.8; 16.5]
Calendar year of birth [%]						
1997	1637 [10.3]	1404 [10.0]	233 [13.2]	-	-	-
1998	8427 [53.2]	7501 [53.3]	926 [52.6]	-	-	-
1999	5779 [36.5]	5178 [36.8]	601 [34.1]	28 [<0.1]	16 [<0.1]	12 [0.1]
2000	-	-	-	1197 [1.2]	841 [0.9]	356 [3.7]
2001	-	-	-	3635 [3.6]	3019 [3.3]	616 [6.4]
2002	-	-	-	8080 [8.1]	6872 [7.6]	1 208 [12.5]
2003	-	-	-	11 833 [11.9]	10 319 [11.4]	1514 [15.6]
2004	-	-	-	12 786 [12.8]	11 360 [12.6]	1426 [14.7]
2005	-	-	-	14 864 [14.9]	13 475 [15.0]	1389 [14.3]
2006	-	-	-	16 411 [16.4]	15 181 [16.8]	1230 [12.7]
2007	-	-	-	15 919 [15.2]	14 144 [15.7]	1055 [10.9]
2008	-	-	-	12 633 [12.6]	11,891 [13.2]	742 [7.7]
2009	-	-	-	3154 [3.2]	3010 [3.3]	144 [1.5]
IBD diagnosis[b] [%]						
IBD	112 [0.7]	99 [0.7]	13 [0.7]	332 [0.3]	285 [0.3]	47 [0.5]
CD	40 [0.3]	36 [0.3]	4 [0.2]	142 [0.1]	126 [0.1]	16 [0.2]
UC	56 [0.4]	50 [0.4]	6 [0.3]	91 [0.1]	75 [0.1]	16 [0.2]
Average maternal smoking level during pregnancy [%]						
No smoking	14 069 [89.3]	14 069 [100]	-	90 128 [90.5]	90 128 [100]	-
1–5 cigarettes/day	644 [4.1]	-	644 [38.3]	6159 [6.2]	-	6159 [65.3]
≥6 cigarettes/day	1036 [6.6]	-	1036 [61.7]	3269 [3.3]	-	3269 [34.7]
Maternal smoking 0–<3 months before pregnancy [%]						
No	-	-	-	60 342 [68.1]	60 179 [75.8]	163 [1.8]
Yes	-	-	-	28 324 [31.9]	19 188 [24.2]	9136 [98.2]
Maternal ETS exposure during pregnancy [%]						
No	11 302 [85.3]	10 240 [86.0]	1062 [78.7]	87 409 [89.2]	81 623 [91.8]	586 [63.9]
Yes	1954 [14.7]	1667 [14.0]	287 [21.3]	10 541 [10.8]	7271 [8.2]	3270 [36.1]
Child ETS exposure by age 12 months [%]	10 674	9694	980	80 869	73 533	7336
No	10 081 [94.4]	9404 [97.0]	677 [69.1]	58 458 [72.3]	57 794 [78.6]	664 [9.1]
Yes	593 [5.6]	290 [3.0]	303 [30.9]	22 411 [27.7]	15 739 [21.4]	6672 [90.9]
Child ETS exposure by age 36 months [%]	8498	7758	740	55 172	50 979	4193
No	7 071 [83.2]	6928 [89.3]	143 [19.3]	47 588 [86.3]	46 558 [91.3]	1030 [24.6]
Yes	1427 [16.8]	830 [10.7]	597 [80.7]	7584 [13.7]	4421 [8.7]	3163 [75.4]
Maternal age at delivery[c] [%]						
<25 years	2469 [15.8]	2024 [14.6]	445 [25.6]	10 925 [11.0]	8694 [9.7]	2231 [23.0]

Table 1. Continued

		ABIS		MoBa		
	All 15 843	No smoking during pregnancy 14 083	Smoking during pregnancy 1 760	All 99 820	No smoking during pregnancy 90 128	Smoking during pregnancy 9 692
25–34 years	11 192 [71.8]	10 146 [73.2]	1046 [60.1]	71 340 [71.5]	65 427 [72.6]	5913 [61.0]
≥35 years	1934 [12.4]	1686 [12.2]	248 [14.3]	17 501 [17.5]	15 957 [17.7]	1544 [16.0]
Parental IBD [%]						
No	15 649 [98.8]	13 905 [98.7]	1744 [99.1]	97 514 [97.7]	88 028 [97.7]	9486 [97.9]
Yes	194 [1.2]	178 [1.3]	16 [0.9]	2306 [2.3]	2100 [2.3]	206 [2.1]
Parental origin[d] [%]						
Norwegian/Swedish	14 098 [89.0]	12 595 [89.7]	1503 [86.2]	86 078 [89.3]	77 917 [89.1]	8161 [92.0]
Other	1690 [10.7]	1449 [10.3]	241 [13.8]	10 262 [10.7]	9548 [10.9]	714 [8.0]
Maternal education level[e] [%]						
Low	1348 [8.5]	880 [6.3]	468 [26.8]	7907 [8.0]	5456 [6.1]	2451 [26.0]
Medium	8772 [55.6]	7739 [55.2]	1033 [59.1]	29 217 [29.5]	24 826 [27.7]	4391 [46.6]
High	5660 [35.9]	5413 [38.6]	247 [14.1]	62 009 [62.6]	59 432 [66.2]	2577 [27.4]

See Supplementary Table 6 and Figure 1 for data missingness.
CD, Crohn's disease; IBD, inflammatory bowel disease; MoBa, the Norwegian Mother, Father, and Child Cohort study; SD, standard deviation; UC, ulcerative colitis; ABIS, All Babies in Southeast Sweden study; ETS, environmental tobacco smoke.
[a]Age at the end of the study equals age at diagnosis for IBD, CD, and UC cases.
[b]IBD also includes IBD-U, inflammatory bowel disease unclassified.
[c]Categories were merged at the presentation, but the following categories were used in the analyses: 15–19/20–24/25–29/30–34/35–39/40–44 years.
[d]Other country if any parent born in a country other than Sweden [ABIS] or any parent having a different native language other than Norwegian [MoBa].
[e]High education level equalled ≥13 years of education, medium education level 12 years of education, and low education level 9–11 years of education [9 years in ABIS; 9–11 years in MoBa].

These children were treated as missing in relevant analyses. Mothers reported ETS exposure during pregnancy at birth in ABIS and at 15 weeks of pregnancy in MoBa, covering ETS exposure at work or home. We categorised maternal ETS exposure during pregnancy into any vs no ETS exposure [yes/no]. Child ETS exposure by 12 [yes/no] and by 36 months of age [yes/no] was defined by household or parental smoking at the corresponding age [Supplementary Table 2].

We also retrieved pregnancy questionnaire data from MoBa considering maternal smoking 0–3 months before pregnancy [yes/no] and paternal smoking during pregnancy. Similar data were not available in ABIS.

Exposure definitions are detailed in Supplementary Table 2.

2.2. Outcomes

Individual-level data in ABIS and MoBa was linked to national patient registers[20,21] through the unique personal identification numbers assigned to all residents at birth or immigration. At least two recorded International Classification of Diseases [ICD]-10 codes defined IBD. The outcome of IBD considered CD, UC, or IBD-U events. We used subtype-specific diagnoses to define the outcome of CD and UC [Supplementary Table 3]. The first recorded ICD code was used to define the time of diagnosis. In ABIS and other Swedish studies, our register-based IBD algorithm has shown a positive predictive value of 93% on medical record reviews.[22] Norwegian data have shown two ICD-10 codes for IBD to have a positive predictive value of 94% for a clinical diagnosis of IBD [unpublished data]. Data on IBD diagnoses were captured throughout December 31, 2020, in ABIS and December 31, 2021, in MoBa.

2.3. Other data

In ABIS, data on the child's sex, parental origin, parental IBD, and maternal education level were retrieved from the questionnaire distributed at birth. Parental origin was categorised into Swedish if both parents were born in Sweden, and into Other if any parent reported another country of birth. Maternal age was retrieved from the Swedish Medical Birth Register.[23] In MoBa, parental origin and maternal education level data were retrieved from pregnancy questionnaires. Parental origin was categorised into Norwegian if both parents recorded Norwegian as their native language, and into Other if any parent reported another native language. Data on the child's sex and maternal age were collected from the Medical Birth Registry of Norway.[24] Parental IBD diagnoses were identified by relevant ICD-10 codes in the Norwegian Patient Register.[20] Data were categorised as shown in Table 1. A recent study, based on the same population as our study, did not find any association between breastfeeding duration and risk of offspring IBD [unpublished data]. Thus, this factor was not adjusted for in our analyses. As previously described,[25] we retrieved information on household income level of children in ABIS and MoBa.

2.4. Statistical analyses

Unadjusted hazard ratios [HRs] and adjusted HRs [aHRs] and 95% confidence intervals [CIs] for IBD, CD, and UC were estimated using Cox proportional hazard regression with age [years] as the time metric. For pregnancy exposures, follow-up started at birth. For child ETS exposures, follow-up began at 12 and 36 months. Follow-up ended at the time of an IBD, CD, or UC diagnosis or was censored at the end of data capture [December 31, 2020, in ABIS and December 31, 2021, in MoBa]. Schoenfeld residuals, interaction with time analyses, and visual data assessment were used to assess the proportional hazard assumption. The assumption was found valid for all analyses, indicating no significant time-dependent associations. Cochran's Q test assessed the between-cohort

heterogeneity.[26] It was found to be non-significant for all pooled analyses of the IBD outcome. A random-effects model, incorporating heterogeneity of effects, was used to pool cohort-specific estimates.[27] Power analysis has been presented in Supplementary Table 4.

All analyses were adjusted for the child's sex, parental IBD and origin, maternal age, and maternal education level,[25] as these were considered potential confounding factors. Statistical analyses were performed using R Statistics Software[28] [version 4.2.2, R packages meta[29] and metaverse[30]] and SPSS Statistics [version 29.0].

2.5. Subanalyses and sensitivity analyses

Pre-planned subanalyses considered the outcome of childhood-onset IBD [ie, <18 years of age]. Finally, we re-ran our analyses after excluding events of IBD diagnoses in children <2.0 years of age, considered a rare and highly genetically determined disease subtype.[31]

Separate sensitivity analyses were performed, further adjusting analyses of maternal smoking in pregnancy and early-life ETS exposure for household income level and paternal smoking in pregnancy [yes vs no] [only available in MoBa]. Finally, analyses for child ETS exposure were also adjusted for maternal smoking during pregnancy.

3. Results

This study included 115 663 children with data on smoking exposure [Figure 1]. During 1 987 430 person-years of follow-up, 444 children were diagnosed with IBD (incidence rate [IR] 31.8/100 000 person-years in ABIS; IR 20.3/100 000 person-years in MoBa; Supplementary Table 5). Mean age at the end of the follow-up was 22.3 years in ABIS and 16.4 years in MoBa [Table 1]. Background characteristics were similar in ABIS and MoBa [Supplementary Table 6] and between those with and without ETS data by 12 and 36 months of follow-up [Supplementary Table 7].

3.1. Early-life smoking exposure and later IBD

Overall, some 10% [11 452/115 663] of the children had a mother who smoked during pregnancy. Maternal smoking during pregnancy [any vs no smoking] yielded a pooled aHR of 1.30 for IBD (95% confidence interval [CI] = 0.97–1.74; Figure 2). A higher level of maternal smoking during pregnancy was associated with an increased risk of IBD [average ≥6 cigarettes per day vs no smoking, pooled aHR = 1.60; 95% CI = 1.08–2.38; average 1–5 cigarettes per day vs no smoking, pooled aHR = 1.09; 95% CI = 0.73–1.64; Figure 2]. A higher level of maternal smoking during pregnancy yielded a higher estimated aHR for UC compared with CD risk [Figure 3–4]. In pooled analyses, maternal ETS exposure during pregnancy was not associated with the risk of developing IBD in the child [any vs no ETS, aHR = 0.88; 95% CI = 0.64–1.21; Figure 2].

Approximately one in four children [25%, 23 004/91 542] were exposed to ETS during their first year. Any vs no ETS exposure in the first year was associated with an increased risk of later IBD [pooled aHR = 1.32;95% CI = 1.03–1.69; Figure 2]. Largely similar but non-significant aHR was found in post hoc analyses of childhood ETS additionally adjusted for maternal smoking during pregnancy [child ETS exposure by 12 months, aHR = 1.25; 95% CI = 0.95–1.65; Supplementary Table 8]. Any vs no ETS exposure in the first year of life yielded a pooled aHR of 1.37 [95% CI = 0.95–1.98] for CD and 1.06 [95% CI = 0.46–2.43] for UC [Figure 3–4]. The child's exposure to ETS by 36 months yielded a pooled aHR of 1.26 for IBD; 95% CI = 0.91–1.73; Figure 2].

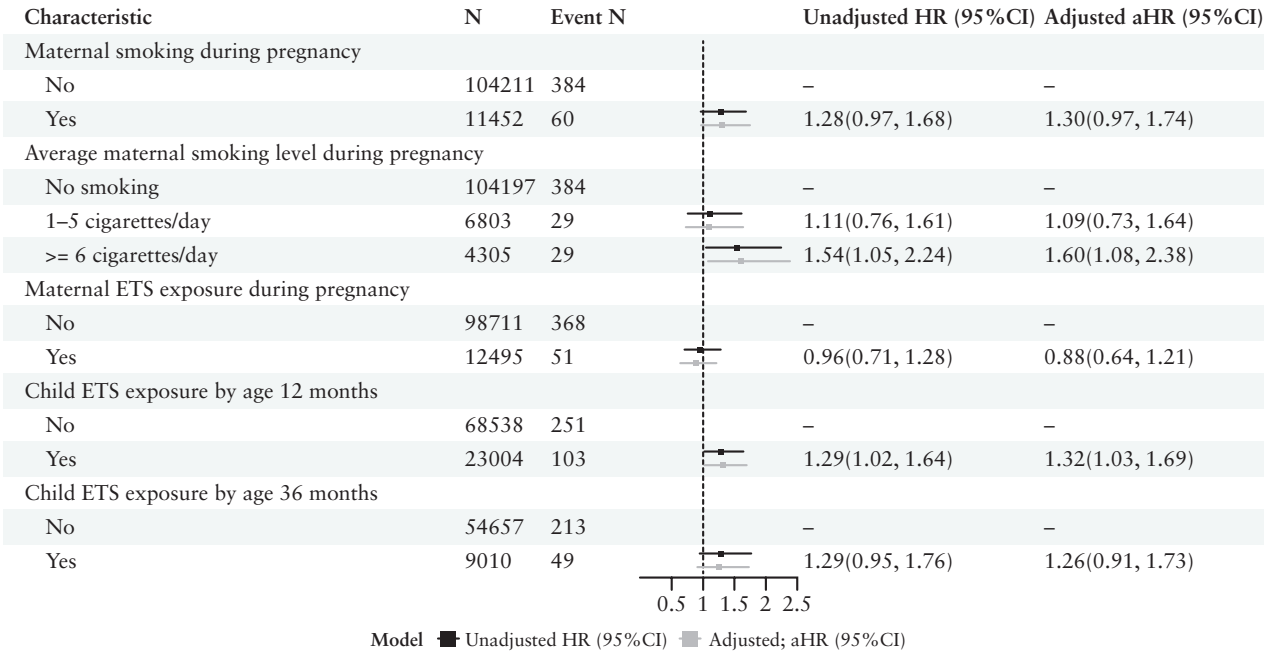


Figure 2 Pooled analyses of early-life smoking exposure and later risk of inflammatory bowel disease [IBD] in the ABIS and MoBa birth cohorts. Adjusted hazard ratio [aHR] for sex, parental IBD, parental origin, maternal age, and maternal education level. CI, confidence interval; ETS, environmental tobacco smoke; HR, hazard ratio; MoBa, the Norwegian Mother, Father, and Child Cohort Study; ABIS, All Babies in Southeast Sweden study.

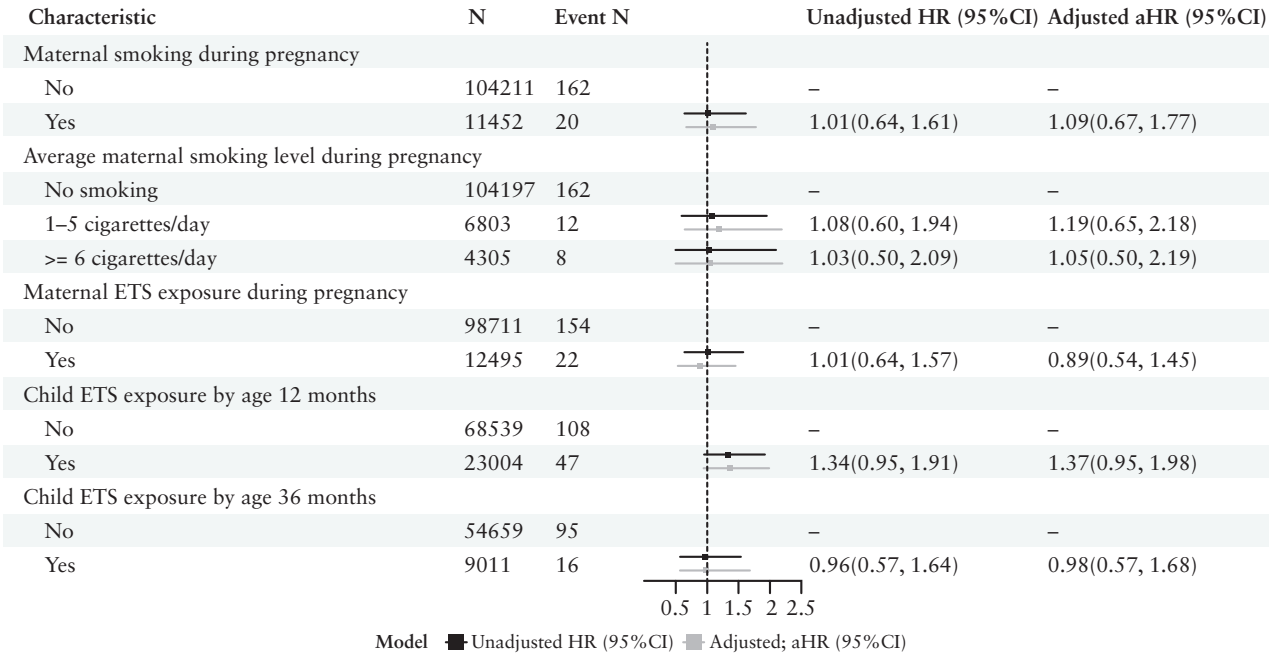


Figure 3 Pooled analyses of early-life smoking exposure and later risk of Crohn's disease in the ABIS and MoBa birth cohorts. Adjusted hazard ratio [aHR] for sex, parental IBD, parental origin, maternal age, and maternal education level. CI, confidence interval; ETS, environmental tobacco smoke; HR, hazard ratio; MoBa, the Norwegian Mother, Father, and Child Cohort Study; ABIS, All Babies in Southeast Sweden study.

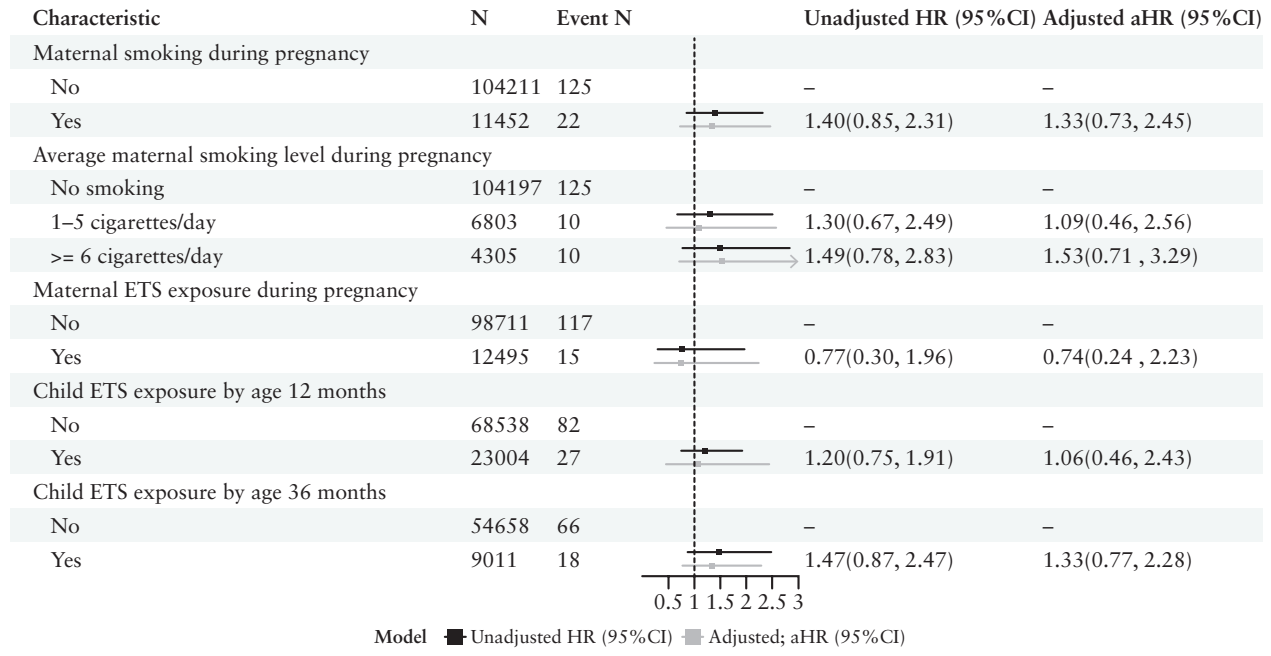


Figure 4 Pooled analyses of early-life smoking exposure and later risk of ulcerative colitis in the ABIS and MoBa birth cohorts. Adjusted hazard ratio [aHR] for sex, parental IBD, parental origin, maternal age, and maternal education level. CI, confidence interval; ETS, environmental tobacco smoke; HR, hazard ratio; MoBa, the Norwegian Mother, Father, and Child Cohort Study; ABIS, All Babies in Southeast Sweden study; IBD, inflammatory bowel disease.

3.2. Subanalyses and sensitivity analyses

Maternal smoking in the last 3 months before pregnancy was not associated with IBD, CD, or UC in the cohort-specific analyses of MoBa [Supplementary Table 9–11].

Restricting the end of follow-up to childhood-onset IBD [<18 years of age] resulted in similar estimates as our main analysis [Supplementary Table 12]. This subanalysis for maternal smoking level during pregnancy and risk of CD was not possible in ABIS due to no cases of CD in one category [data not shown]. Excluding three individuals [all in MoBa] with an IBD diagnosis before the age of 2 years yielded similar estimates to those of the main analysis [data not shown].

Seperate sensitivity analyses further adjusted for [i] household income level, and [ii] paternal smoking in pregnancy

[only available in MoBa]; all yielded essentially unchanged results [Supplementary Tables 13 and 14, respectively].

4. Discussion

This Scandinavian birth cohort study of over 115, children examined the association between early-life smoking exposure and the child's risk of later developing IBD. We found that a higher level of maternal smoking during pregnancy, independent of examined confounding variables, increased the risk of IBD in the child. Child ETS exposure in the first year of life was also associated with an increased risk of IBD, with similar albeit non-significant HRs after adjustment for maternal smoking during pregnancy. On the other hand, neither maternal ETS exposure during pregnancy nor maternal smoking before pregnancny, was linked to offspring IBD.

4.1. Earlier research

A recent meta-analysis, based on retrospective and prospective data, showed that maternal smoking during pregnancy was associated with IBD (odds ratio [OR] = 1.49; 95% CI = 1.17–1.90), although no associations were found in analyses according to IBD subtype.[7] However, there are some inconsistencies across earlier publications [Supplementary Table 1].

A US study of 253 patients with CD and 177 patients with UC investigated childhood ETS exposure as reported by the patients in adulthood [CD patients: mean age 45 years; UC patients: mean age 47 years]. Using these retrospectively collected data, they found maternal smoking during pregnancy to be more common in individuals who developed CD vs healthy controls [CD: OR = 1.72; 95% CI = 1.10–2.71; UC: OR = 1.53; 95% CI = 0.93–2.49].[9]

In contrast, using Swedish register-based data of 452 children [<16 years] with hospital admission for IBD, Aspberg *et al.* found that maternal smoking early during pregnancy was associated with a decreased risk of IBD [OR = 0.71; 95% CI = 0.55–0.91].[8] This study differs from ours in several ways. First, Aspberg *et al.* were limited to inpatients with IBD and therefore more prone to include IBD patients with severe phenotype or comorbidities. Second, whereas our data were captured between 1997 and 2009, Aspberg *et al*. retrieved smoking data over 1987–2000. This difference could influence the results, given the significant changes in smoking habits. Additionally, only smoking in early pregnancy was assessed,[8] which can be compared with any time of pregnancy evaluated in our study. More recently, based on prospectively collected register data in Denmark, Spangmose *et al*. did not find any association between maternal smoking during pregnancy and paediatric CD or UC [CD, aHR = 0.92; 95% CI = 0.74–1.15; UC, aHR = 0.93; 95% CI = 0.74–1.17].[10] However, these authors acknowledge a high degree of missingness for smoking data which may have hampered their ability to identify an association.

In addition to the prospectively conducted studies described above,[8,10] retrospective case-control studies have also shown inconsistent results [Supplementary Table 1]. Whereas most studies have found no significant associations between prenatal or childhood smoking exposure and later IBD,[32–37] some studies have presented positive associations linking smoking exposure to an increased risk of offspring IBD.[38–40] These discrepancies may be due to several factors, including differences in the study populations, years under study, diagnostic criteria, and definitions of smoking exposure.

4.2. Explanation of findings

We found that a higher level of maternal smoking was associated with IBD in the child, rather than any smoking exposure during pregnancy. We lacked detailed information about the duration of any smoking during pregnancy, but it is likely that such use was limited to the very first weeks before confirmed pregnancy. Thus in this study, the variable for the average level of smoking during pregnancy could be a better proxy for tobacco smoke exposure of the fetus/child compared with any vs no smoking, as it reflects the mean smoking exposure over the entire pregnancy, independent of when smoking occurred during the pregnancy. Still, we acknowledge that our 95% CIs for the associations of any maternal smoking in pregnancy with offspring IBD [aHR = 1.30; 95% CI = 0.97–1.74] do not rule out small-to-modest associations. Moreover, given our sample size and the distribution of maternal smoking exposure during pregnancy, neither extreme exposure levels of maternal smoking nor dose-response relationships could be further examined. Less than 10% of the total number of events was distributed in the highest level of smoking category. Our moderate smoking exposure likely reflects a generally low and decreasing smoking exposure of Norwegian and Swedish women during pregnancy.[41,42]

The association in our study between a high number of cigarettes smoked during pregnancy and the risk of IBD may reflect the risk estimates for UC rather than for CD. This finding may mirror the results of adult former smokers having an increased risk of UC.[6] However, given the tendency to change types of IBD during follow-up, we urge caution in interpreting subtype-specific results. The biological mechanisms underlying this phenomenon are poorly understood. The children in our study were exposed to tobacco smoke at a young age, and we do not know whether there is a continuation of smoking exposure or whether they could be seen to some extent as 'ex-smokers'. We also cannot rule out the possibility that participants exposed to smoking in early life are more likely to start smoking themselves later in life, affecting the risk of IBD. Smoking is a direct risk of CD in adults[5] but we did not measure continuous exposure to smoking [indirect or direct].

If it is a causal association, exactly which components in cigarette smoke initiate pathological pathways to IBD are not fully understood. Current and former smokers have reduced gut bacterial diversity.[43] Infancy is a critical period for microbiota development, and the environment has a vital impact on microbial growth.[44] Fetal exposure to tobacco smoke through the mother during pregnancy[13] or ETS exposure in early life likely contributes to alterations in the child's gut microbiota. Also, reactive oxygen species and other components leading to oxidative stress might negatively affect the tissue of the gastrointestinal tract,[6] and nicotine has been suggested as an immunosuppressive component.[45] Additionally, epigenetic changes might occur due to prenatal smoking exposure.[46]

4.3. Strengths and limitations

This study has several strengths. Using two similar large-scale prospective cohorts allowed us to provide robust risk estimates for IBD and to examine the consistency across cohort-specific results. We used questionnaire data previously validated against the measurement of cotinine, a biomarker for nicotine exposure.[16–19] In ABIS, parental smoking behaviour by self-reported questionnaire data significantly

affected urine cotinine levels in children.[16] In MoBa, self-reported daily smoking, validated against plasma cotinine levels, showed a sensitivity of 82% and a specificity of 99%.[17] This validation is important, given that self-reported smoking data may be influenced by social desirability bias, ie, a tendency to under-report smoking during pregnancy because of the associated stigma. Other studies have shown that self-reported smoking data may be more valid than practitioner register-based.[47] Identification of IBD was retrieved from two high-quality national patient registers in both cohorts,[20,21,48] with a minimum of two diagnostic records of IBD, highly consistent with a clinical diagnosis of IBD in both Sweden and Norway (positive predictive value [PPV] ≥93%).[22,49] Similarities in PPV for IBD in Norway and Sweden are expected, given the similar health care systems and health care register data of Nordic countries.[20,50] Another strength is our use of comprehensive covariate data to examine whether other lifestyle characteristics may confound any associations observed. For example, we adjusted for maternal education level, a predictor of smoking during pregnancy and linked to IBD.[51,52]

However, we acknowledge some study limitations. The two included cohorts differed in the participation rate [ABIS, 79%; MoBa, 41%], possibly related to the regional approach in ABIS vs the nationwide approach in MoBa. Compared with all women giving birth in Norway during the same period, women in the MoBa cohort were older and less likely to smoke.[53] These differences have been shown not to significantly affect exposure-outcome associations.[53] This self-selection may indicate a reduced generalisability to under-represented populations. Additionally, we used data from two developed countries with a high IBD incidence.[54,55] Our IBD subtype-specific analyses failed to identify any statistically significant associations similar to the findings for IBD overall, possibly because of the few IBD cases present in some examined groups. We were also unable to examine whether early-life smoking exposure may influence IBD phenotypes or disease severity. Due to the shorter follow-up time in MoBa, we might have failed to capture IBD cases that would develop later in the children.

As in any observational setting, we could not account for unmeasured or residual confounding. For example, besides maternal education level, we did not have data on complete socioeconomic characteristics, such as socio-cultural factors that might be associated with smoking. Futher, smoking exposure often reflects a family's overall health behaviour, including dietary habits and living conditions, which may influence the risk of IBD.

In conclusion, this prospective Scandinavian cohort study found an association between a higher level of maternal smoking during pregnancy and ETS exposure in the first year of life and subsequent risk of IBD.

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Conflict of Interest

The authors declare no conflict of interest.

Specific author contributions

Designed the study: IS, KS, JL, and KM. Collected data: KS and JL. Analysed the data: IS, BA, KM. Wrote the first draft of the paper: IS and KM. Contributed to the study design, interpretation of data, and writing: all authors. Interpretation of data and approval of final version: all authors. Responsible for data integrity: KM. Obtained funding for the analyses of this project: KM. Guarantor of the article: KM.

Data Availability

The data underlying this article cannot be shared publicly due to legal and ethical regulations in Norway and Sweden.

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Ethical Considerations

ABIS was approved by the Research Ethics Committees of the Faculty of Health Sciences, Linköping University [Li 287-96] and the Medical Faculty of Lund University [Lu 83-97], with ethical approvals Dnr: 96287; 03-092; 03-513; 2018/380-32; 1849/2021. All participating mothers in the ABIS cohort gave written informed consent to participate. The establishment of MoBa and initial data collection were based on a licence from the Norwegian Data Protection Agency and approval from the Regional Committees for Medical and Health Research Ethics [no. 153328]. All participanting parents in the MoBa cohort gave written informed consent to participate.

Supplementary Data

Supplementary data are available at *ECCO-JCC* online.

References

1. Kuenzig ME, Fung SG, Marderfeld L, *et al*.; InsightScope Pediatric IBD Epidemiology Group. Twenty-first century trends in the global epidemiology of pediatric-onset inflammatory bowel disease: systematic review. *Gastroenterology* 2022;**162**:1147–59.e4.
2. Roberts SE, Thorne K, Thapar N, *et al*. A systematic review and meta-analysis of paediatric inflammatory bowel disease incidence and prevalence across Europe. *J Crohns Colitis* 2020;**14**:1119–48.
3. Zhao M, Gönczi L, Lakatos PL, Burisch J. The burden of inflammatory bowel disease in Europe in 2020. *J Crohns Colitis* 2021;**15**:1573–87.

4. Ananthakrishnan AN. Epidemiology and risk factors for IBD. *Nat Rev Gastroenterol Hepatol* 2015;**12**:205–17.
5. Piovani D, Danese S, Peyrin-Biroulet L, Nikolopoulos GK, Lytras T, Bonovas S. Environmental risk factors for inflammatory bowel diseases: an umbrella review of meta-analyses. *Gastroenterology* 2019;**157**:647–59.e4.
6. Berkowitz L, Schultz BM, Salazar GA, *et al*. Impact of cigarette smoking on the gastrointestinal tract inflammation: opposing effects in Crohn's disease and ulcerative colitis. *Front Immunol* 2018;**9**:74.
7. Agrawal M, Sabino J, Frias-Gomes C, *et al*. Early life exposures and the risk of inflammatory bowel disease: systematic review and meta-analyses. *EClinicalMedicine* 2021;**36**:100884.
8. Aspberg S, Dahlquist G, Kahan T, Källén B. Fetal and perinatal risk factors for inflammatory bowel disease. *Acta Paediatr* 2006;**95**:1001–4.
9. Mahid SS, Minor KS, Stromberg AJ, Galandiuk S. Active and passive smoking in childhood is related to the development of inflammatory bowel disease. *Inflamm Bowel Dis* 2007;**13**:431–8.
10. Spangmose AL, Jørgensen MH, Jakobsen C, *et al*. Pre- and perinatal exposures associated with developing pediatric-onset immune-mediated inflammatory disease: a Danish nation-wide cohort study. *J Autoimmun* 2023;**136**:103032.
11. Stämpfli MR, Anderson GP. How cigarette smoke skews immune responses to promote infection, lung disease and cancer. *Nat Rev Immunol* 2009;**9**:377–84.
12. Li LF, Chan RL, Lu L, *et al*. Cigarette smoking and gastrointestinal diseases: the causal relationship and underlying molecular mechanisms [review]. *Int J Mol Med* 2014;**34**:372–80.
13. Munyaka PM, Khafipour E, Ghia JE. External influence of early childhood establishment of gut microbiota and subsequent health implications. *Front Pediatr* 2014;**2**:109.
14. Joubert BR, Håberg SE, Bell DA, *et al*. Maternal smoking and DNA methylation in newborns: in utero effect or epigenetic inheritance? *Cancer Epidemiol Biomarkers Prev* 2014;**23**:1007–17.
15. Magnus P, Birke C, Vejrup K, *et al*. Cohort Profile Update: The Norwegian Mother and Child Cohort Study [MoBa]. *Int J Epidemiol* 2016;**45**:382–8.
16. Johansson A, Hermansson G, Ludvigsson J. How should parents protect their children from environmental tobacco-smoke exposure in the home? *Pediatrics* 2004;**113**:e291–5.
17. Kvalvik LG, Nilsen RM, Skjærven R, *et al*. Self-reported smoking status and plasma cotinine concentrations among pregnant women in the Norwegian Mother and Child Cohort Study. *Pediatr Res* 2012;**72**:101–7.
18. Mårild K, Tapia G, Midttun O, *et al*. Smoking in pregnancy, cord blood cotinine and risk of celiac disease diagnosis in offspring. *Eur J Epidemiol* 2019;**34**:637–49.
19. Reese SE, Zhao S, Wu MC, *et al*. DNA methylation score as a biomarker in newborns for sustained maternal smoking during pregnancy. *Environ Health Perspect* 2017;**125**:760–6.
20. Bakken IJ, Ariansen AMS, Knudsen GP, Johansen KI, Vollset SE. The Norwegian patient registry and the Norwegian registry for primary health care: research potential of two nationwide health-care registries. *Scand J Public Health* 2020;**48**:49–55.
21. National Board of Health and Welfare. *Information Available in the National Patient Register [NPR]* 2016. Accessed September 15, 2023. https://www.socialstyrelsen.se/globalassets/sharepoint-dokument/dokument-webb/statistik/information-in-the-national-patient-register.pdf
22. Mouratidou N, Malmborg P, Järås J, *et al*.; SWIBREG Study Group. Identification of childhood-onset inflammatory bowel disease in Swedish healthcare registers: a validation study. *Clin Epidemiol* 2022;**14**:591–600.
23. Källén B, Källén K, Olausson OP. *The Swedish Medical Birth Register - a Summary of Content and Quality. Swedish National Board of Health and Welfare,* Centre for Epidemiology; 2003. Accessed December 20, 2022. https://www.socialstyrelsen.se/globalassets/sharepoint-dokument/artikelkatalog/ovrigt/2003-112-3_20031123.pdf
24. Irgens LM. The Medical Birth Registry of Norway. Epidemiological research and surveillance throughout 30 years. *Acta Obstet Gynecol Scand* 2000;**79**:435–9.
25. Sigvardsson I, Størdal K, Östensson M, Guo A, Ludvigsson J, Mårild K. Childhood socioeconomic characteristics and risk of inflammatory bowel disease: a Scandinavian birth cohort study. *Inflamm Bowel Dis* 2023 Sep 22:izad220.
26. Cochran WG. Some methods for strengthening the common χ 2 tests. *Biometrics* 1954;**10**:417.
27. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials* 1986;**7**:177–88.
28. R Core Team. *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing; 2022.
29. Balduzzi S, Rücker G, Schwarzer G. How to perform a meta-analysis with R: a practical tutorial. *Evid Based Ment Health* 2019;**22**:153–60.
30. Viechtbauer W. Conducting meta-analyses in R with the metafor Package. *J Stat Softw* 2010;**36**:1–48.
31. Kappelman MD, Grand RJ. Does inflammatory bowel disease develop in infants? *Inflamm Bowel Dis* 2008;**14**:S6–8.
32. Abdallah B, Arif M, Al-Malki M, *et al*. The association between inflammatory bowel disease and exposure to tobacco smoking: a case-control study in Qatar. *Int J Gen Med* 2023;**16**:233–42.
33. Baron S, Turck D, Leplat C, *et al*. Environmental risk factors in paediatric inflammatory bowel diseases: a population based case control study. *Gut* 2005;**54**:357–63.
34. Han DY, Fraser AG, Dryland P, Ferguson LR. Environmental factors in the development of chronic inflammation: a case-control study on risk factors for Crohn's disease within New Zealand. *Mutat Res* 2010;**690**:116–22.
35. Persson PG, Ahlbom A, Hellers G. Inflammatory bowel disease and tobacco smoke: a case-control study. *Gut* 1990;**31**:1377–81.
36. Sandler RS, Sandler DP, McDonnell CW, Wurzelmann JI. Childhood exposure to environmental tobacco smoke and the risk of ulcerative colitis. *Am J Epidemiol* 1992;**135**:603–8.
37. Uchiyama K, Haruyama Y, Shiraishi H, *et al*. Association between Passive Smoking from the Mother and Pediatric Crohn's Disease: a Japanese Multicenter Study. *Int J Environ Res Public Health* 2020;**17**(8): 2926.
38. Basson A, Swart R, Jordaan E, Mazinu M, Watermeyer G. The association between childhood environmental exposures and the subsequent development of Crohn's disease in the Western Cape, South Africa. *PLoS One* 2014;**9**:e115492.
39. Gearry RB, Richardson AK, Frampton CM, Dodgshun AJ, Barclay ML. Population-based cases control study of inflammatory bowel disease risk factors. *J Gastroenterol Hepatol* 2010;**25**:325–33.
40. Lashner BA, Shaheen NJ, Hanauer SB, Kirschner BS. Passive smoking is associated with an increased risk of developing inflammatory bowel disease in children. *Am J Gastroenterol* 1993;**88**:356–9.
41. Kvalvik LG, Skjaerven R, Haug K. Smoking during pregnancy from 1999 to 2004: a study from the Medical Birth Registry of Norway. *Acta Obstet Gynecol Scand* 2008;**87**:280–5.
42. Reitan T, Callinan S. Changes in smoking rates among pregnant women and the general female population in Australia, Finland, Norway, and Sweden. *Nicotine Tob Res* 2017;**19**:282–9.
43. Shanahan ER, Shah A, Koloski N, *et al*. Influence of cigarette smoking on the human duodenal mucosa-associated microbiota. *Microbiome* 2018;**6**:150.
44. Li M, Wang M, Donovan SM. Early development of the gut microbiome and immune-mediated childhood disorders. *Semin Reprod Med* 2014;**32**:74–86.
45. Sopori M. Effects of cigarette smoke on the immune system. *Nat Rev Immunol* 2002;**2**:372–7.
46. Fuemmeler BF, Dozmorov MG, Do EK, *et al*. DNA methylation in babies born to nonsmoking mothers exposed to secondhand smoke during pregnancy: an epigenome-wide association study. *Environ Health Perspect* 2021;**129**:57010.

47. Mant J, Murphy M, Rose P, Vessey M. The accuracy of general practitioner records of smoking and alcohol use: comparison with patient questionnaires. *J Public Health Med* 2000;**22**:198–201.
48. Laugesen K, Ludvigsson JF, Schmidt M, *et al*. Nordic health registry-based research: a review of health care systems and key registries. *Clin Epidemiol* 2021;**13**:533–54.
49. Östensson M, Björkqvist O, Guo A, *et al*. Epidemiology, validation, and clinical characteristics of inflammatory bowel disease: the ABIS birth cohort study. *BMC Gastroenterol* 2023;**23**:199.
50. Ludvigsson JF, Andersson E, Ekbom A, *et al*. External review and validation of the Swedish national inpatient register. *BMC Public Health* 2011;**11**:450.
51. Härkönen J, Lindberg M, Karlsson L, Karlsson H, Scheinin NM. Education is the strongest socio-economic predictor of smoking in pregnancy. *Addiction* 2018;**113**:1117–26.
52. Do EK, Green TL, Prom-Wormley EC, Fuemmeler BF. Social determinants of smoke exposure during pregnancy: findings from waves 1 & 2 of the Population Assessment of Tobacco and Health [PATH] Study. *Prev Med Rep* 2018;**12**:312–20.
53. Nilsen RM, Vollset SE, Gjessing HK, *et al*. Self-selection and bias in a large prospective pregnancy cohort in Norway. *Paediatr Perinat Epidemiol* 2009;**23**:597–608.
54. Forss A, Clements M, Bergman D, *et al*. A nationwide cohort study of the incidence of inflammatory bowel disease in Sweden from 1990 to 2014. *Aliment Pharmacol Ther* 2022;**55**:691–9.
55. Lirhus SS, Høivik ML, Moum B, Anisdahl K, Melberg HO. Incidence and prevalence of inflammatory bowel disease in Norway and the impact of different case definitions: a nationwide registry study. *Clin Epidemiol* 2021;**13**:287–94.